

Training-time electrical elimination in physics-informed neural networks for electrothermal phase reconstruction

Authors: [author names and affiliations to be supplied]

Corresponding author: [name and email to be supplied]

Abstract

A final electrical solve can repair terminal outputs while leaving inaccurate temperature and phase unchanged. We study how this constraint should participate in learning a transient electrothermal state from prescribed field observations. A partially eliminated PINN retains neural temperature and phase, solves a grounded finite-volume electrical problem during training, and uses its face resistances for consistent local Joule deposition. The soft PINN and interpolation controls receive the same final electrical solve. Four paired fits on two separately reconstructed finite-pulse protocols retain the complete device-advantage criterion under three numerical references and two electrical readers. With the finer reader and spatial reference, current and power error reductions over the soft controls are 47.03-76.52% and 47.86-77.50%. Strong interpolation gives a continuous-ranking counterexample. Clean full-label controls without the thermal/phase interior residuals have lower phase, current and power errors. A further frozen experiment removes phase labels throughout the second cycle while retaining potential, temperature and post-window phase observations. Neither initialization establishes the complete phase-gap criterion against D_E under the three references and two readers. Outside-window noninferiority costs occur in 6 of the twelve primary comparison conditions. The affected quantities are top-current NRMSE, potential RMS, bottom-current NRMSE, power-trace NRMSE, with every condition retained in the comparison table. The evidence distinguishes the tested electrical-constraint configuration from the test of added dynamic-residual value. It concerns a synthetic, known-model offline reconstruction problem; strict-event, numerical-discretization and material-validation limits remain explicit.

Keywords: physics-informed neural network; electrothermal coupling; electrical elimination; missing phase observations; phase-field reconstruction

1. Problem and information conditions

Small potential errors can produce large contact-current and local-heating errors in a coupled phase-transition calculation. Electrical repair can restore terminal consistency without repairing the learned evolving state. We therefore ask two separate questions: whether training with the electrical constraint improves reconstruction after every method receives the same final solve, and whether the additional thermal/phase interior residuals contribute under complete and phase-gap observation conditions.

PINNs combine observations with explicit differential-equation residuals [1]. We study how electrical constraints participate in reconstructing transient temperature and phase from prescribed field observations. The task is an offline, known-model reconstruction benchmark: geometry, coefficients, initial and boundary conditions, and the complete drive are supplied. Each of the two protocols is fitted separately. Its original observation set

contains 28,875 positive-time locations and 86,625 noiseless scalar labels for potential, temperature and phase. These locations constitute 1.8046875% of the $3,200 \times 500$ positive-time carrier positions; at each observation time, 231 spatial locations cover 7.21875% of the carrier cells. Noiseless labels still inherit the carrier's numerical discretization error. These experiments do not establish a minimal sensor arrangement, zero-shot transfer or calibrated material reconstruction.

The contribution is a specific coupling interface and a matched test of its reconstruction benefit. Neural temperature and phase determine a grounded finite-volume electrical solution. The same face resistances determine local Joule deposition in the thermal balance. Providing the same final electrical solution to the soft and interpolation controls tests whether final electrical repair explains the measured differences. The existing experiments support this training configuration relative to the tested soft comparator. They do not establish a new implicit-differentiation principle or an independent benefit from the additional thermal/phase interior residuals.

Differentiable constrained learning already includes solver-in-the-loop correction [3], hybrid FEM-NN models [4], PDE-constrained layers [15], and modular implicit differentiation [5]. Related approaches enforce hard boundary maps or constrained optimization [16], prescribed integral targets [12], differential-algebraic constraints [17], and local discrete projections [18]. Here the transient T/ϕ states remain neural, while the full grounded electrical subproblem is eliminated. This identifies a physical interface adaptation and its controlled comparison; it does not imply general superiority over all penalty or constrained-optimization methods.

Reduced and all-at-once PDE formulations provide the broader setting for eliminating a constrained variable versus jointly optimizing all fields [13]. Here only quasi-static potential is eliminated; the transient states remain neural.

The field network follows the smooth modified-MLP construction of Wang et al. [2]. The spatial cell-balance thermal residual is related to control-volume learning [6], while the nonconserved phase dynamics use an Allen-Cahn-type potential [7]. The device layout and electrothermal feedback are inspired by wall-cell phase-change-memory modeling [8]; its Ge-rich GST material formulation is distinct from the present dimensionless single-phase variable. The parameters below define the synthetic benchmark and are not an oxide calibration.

2. Two-dimensional physical object

2.1 Equations, geometry and known coefficients

The computational cell is $\Omega = [-1,1] \times [0,1]$, with coordinates x and z and time $0 \leq t \leq 2.5$. Potential V , reduced temperature T and phase fraction ϕ obey

$$\nabla \cdot (\sigma(T, \phi) \nabla V) = 0. \quad (1)$$

$$T_t + L\phi_t = \alpha \Delta T - \gamma T + Q\sigma(T, \phi) |\nabla V|^2. \quad (2)$$

$$\phi_t = M(T) [\varepsilon^2 \Delta \phi - \partial_\phi W(\phi, T)]. \quad (3)$$

The conductivity, mobility and local phase potential are

$$\sigma = \exp[\beta T + \log(R)\phi^2(3 - 2\phi)], \quad (4)$$

$$M(T) = m_c + (m_h - m_c)\text{sigmoid}[(T - T_c)/w_M], \quad (5)$$

$$W = B\phi^2(1 - \phi)^2 + D(T_c - T)\phi^2(3 - 2\phi). \quad (6)$$

We use $\alpha = 0.10$, $\gamma = 4$, $L = 0.05$, $Q = 4$, $\beta = 0.25$, $R = 8$, $\varepsilon = 0.04$, $B = 1$, $D = 6$, $T_c = 0.45$, $m_c = 0.5$, $m_h = 5$ and $w_M = 0.08$. Mobility multiplies the entire phase bracket; it is not moved inside a divergence. Because W depends on the externally evolving temperature, the model is not assigned an autonomous monotone free-energy constraint.

The full top electrode is driven by $U(t)$; a centered bottom segment $|x| \leq 0.35$ is grounded. All remaining electrical faces are insulating. The top temperature is zero; the other thermal boundaries satisfy $\partial_n T + 0.25T = 0$. Phase satisfies $\partial_n \phi = 0$ on every boundary. Initially $T = 0$ and

$$\phi_0 = 0.02 + 0.01\exp\left[-\frac{1}{2}((x/0.18)^2 + ((z - 0.12)/0.10)^2)\right]. \quad (7)$$

These assumptions define a synthetic wall-cell benchmark. No dimensional material parameters, experimental memory operation, or oxide-specific constitutive identification are inferred from it.

2.2 Initial-boundary interpretation

For the initial phase in Eq. (7), at $x=0$ on the bottom boundary, its outward normal derivative is $-0.12 \exp(-0.72) = -0.0584102707151966$. It is therefore not pointwise compatible with the phase Neumann condition at $t=0$. The reference applies the stated discrete no-flux operator during subsequent evolution; the neural representation retains the exact initial map and penalizes phase boundary residuals. We do not assume classical smooth compatibility on the closed space-time boundary.

Direct inspection of the twelve saved historical full-label calibration/L-BFGS/audit pools for the two protocols and two seeds finds no $t=0$ boundary node and hence zero boundary quadrature mass there. This finite-pool fact is distinct from a probabilistic argument about continuous random draws. The existing Adam rule generates boundary times from the four physical windows without explicitly inserting $t=0$; it does not mathematically exclude a zero random variate. The compatibility limitation is retained regardless. Its effect on the E/D_E ranking is unknown; neither the initial state nor the boundary rule has been changed.

2.3 Pulse histories and observations

Each pulse rises linearly from zero to 0.72 over its first 0.05 time units, holds until local time 0.27, then falls to zero at 0.35. The original protocol starts two pulses at 0 and 1.25. The intervention changes only the second start to 1.01, leaving the pulse shape, total time, coefficients and boundaries fixed. The input contains exactly two pulses; it is not periodically extended to create a third pulse. Advancing the second pulse by 0.24 preserves its phase relative to the 0.02 observation spacing.

Support trajectories use 80×40 cells, a main time step of 0.0025 and saving every two steps. The spatial and temporal masks take indices 0,4,8,... and the last index, fixed independently of field values. The resulting $21 \times 11 \times 126$ mask contains 231 analytic initial positions and 28,875 positive-time positions, with 86,625 scalar field labels. Training receives only these samples, coordinates and known physical laws. Dense support fields, reference fields, reference currents or heating, and sealed stress cases do not enter training or model selection.

The evaluation reference uses 160×80 cells, a main time step of 0.000625 and saving every four steps, giving 1001 times. Both numerical trajectories use backward-Euler finite-volume steps with converged electrical/phase/thermal block iterations and logit Newton for phase, as specified in Supplement S1.1. Their algebraic convergence tolerances do not certify discretization accuracy. The grids serve different roles; their existence alone is not a convergence study. Each protocol has its own observations and is fitted separately. The intervention is consequently a complete-case adaptation experiment, not zero-shot transfer or a formal out-of-distribution test. The two seed values are reused across protocols for paired reporting, not counted as four independent physical cases.

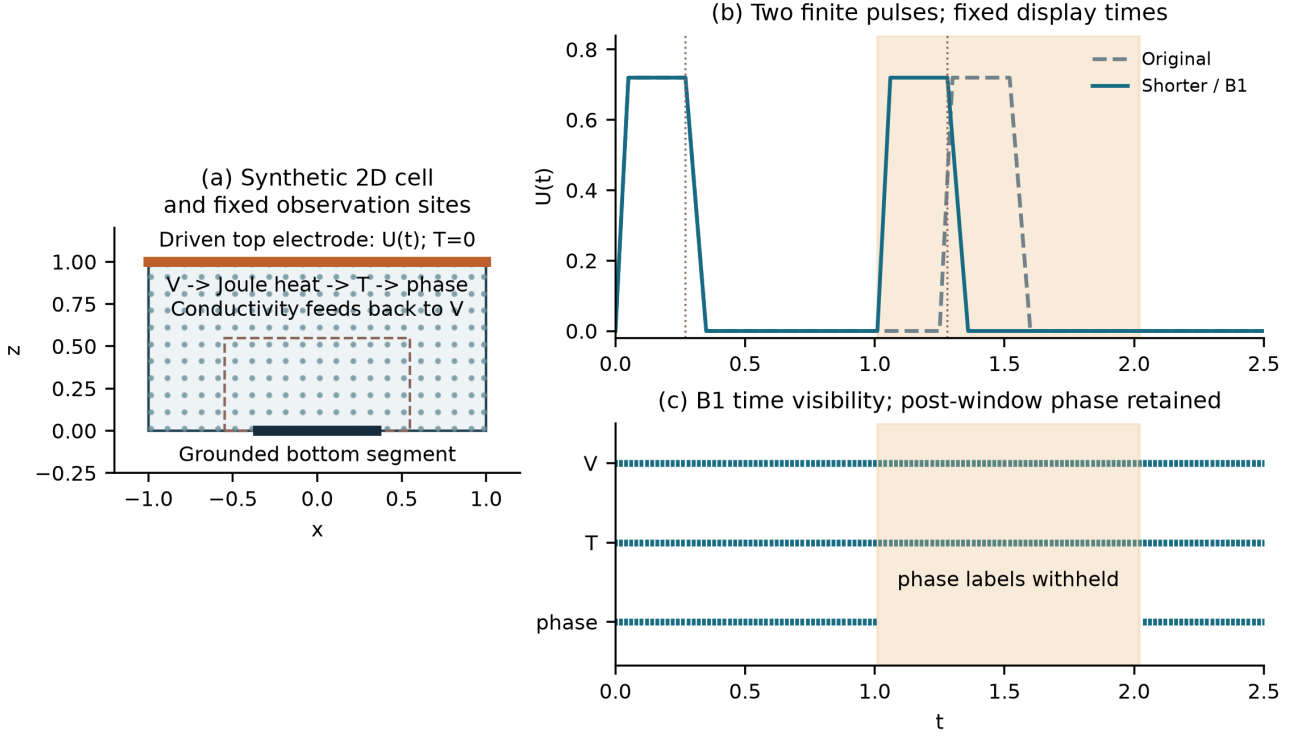


Figure 1. Actual geometry, fixed spatial sampling and finite pulse protocols. The shaded interval $W=[1.01, 2.02]$ removes only phase labels in B1; potential and temperature remain observed. Phase observations after W remain available. Dashed vertical lines mark the predeclared display times 0.27 and 1.28. The dashed spatial box is the fixed state-error ROI.

3. Method and matched controls

3.1 Neural states and electrical elimination

Let $u_\theta = (T_\theta, \phi_\theta)$. Independent smooth modified-MLP heads [2] have width 64 and four hidden layers; the shared temperature architecture also includes a small additive latent adapter described in the Supplement. FP64 arithmetic and identical output mappings are used in both methods. With $a(t) = 1 - \exp(-t/0.35)$,

$$T_\theta = 2.5a(t)(1 - z)\text{sigmoid}(h_T), \quad (8)$$

$$\phi_\theta = \text{sigmoid}[\text{logit}(\phi_0) + 8a(t)h_\phi]. \quad (9)$$

The eliminated method, E, evaluates σ_θ from these states and solves the grounded electrical system

$$A(\sigma_\theta)v^* = f(U, \sigma_\theta). \quad (10)$$

For an internal face e between cells i and j , the half-cell distances d and face area A_e define

$$R_{ei} = d_{ei}/(\sigma_i A_e), \quad R_{ej} = d_{ej}/(\sigma_j A_e), \quad g_e = (R_{ei} + R_{ej})^{-1}. \quad (11)$$

Electrode faces use the corresponding half-cell resistance and the actual heater overlap. Insulating faces contribute no outward conductance. Positive conductances on the connected grid, with Dirichlet electrodes fixing its gauge, make the assembled electrical matrix nonsingular. Electrical enforcement means solving this discrete system to numerical precision; it is not an exact continuum solution. The implementation uses sparse factorization, not a dense inverse. For a scalar loss, the complete first-order derivative is obtained from

$$A^T z = \partial \mathcal{L} / \partial v, \quad d\mathcal{L} = d\mathcal{L}_{\text{direct}} + z^T(df - dA v^*). \quad (12)$$

The direct term includes the explicit dependence of heating on conductivity. The right-hand side f also depends on electrode conductance, and its derivative is retained. A factorization may be reused for the adjoint of the same forward solve, but not after changing the state. This is an application of implicit differentiation [5], with no claim of a new differentiation theorem or support for arbitrary second-order derivatives.

3.2 Consistent local Joule deposition and remaining residuals

Given either the solved or neural potential, face current and deposition are

$$I_e = g_e(v_i - v_j), \quad P_{e \rightarrow i} = I_e^2 R_{ei}, \quad P_{e \rightarrow j} = I_e^2 R_{ej}. \quad (13)$$

Electrode power enters the neighboring boundary cell. For cell volume ω_i , q_i is its deposited power divided by ω_i . This allocation is not generally an equal split between adjacent cells. It gives

$$\sum_i \omega_i q_i = P_J, \quad (14)$$

where P_J sums all internal and electrode dissipation. Equation (14) is an accounting identity for the shared face network, even when a neural potential violates electrical balance. Equality with terminal power UI additionally requires the electrical equations. Neither identity by itself establishes accuracy relative to the reference heating field. P_J denotes electrical dissipation; the reduced thermal source is Qq , so its integral is QP_J . This normalization and the temperature-dependent phase potential do not imply a general total-free-energy decay theorem.

The temperature residual uses a cell balance,

$$R_{T,i} = \partial_t(\bar{T}_i + L\bar{\phi}_i) + \gamma\bar{T}_i - \frac{\alpha}{\omega_i} \sum_f A_f (\nabla T_\theta)_f \cdot n_{if} - Qq_i. \quad (15)$$

Cell means use midpoint quadrature. Temperature gradients are differentiated at shared face nodes with opposite normals for neighboring cells, including the model's own gradients on boundary faces. The same thermal boundary residual remains in both losses. The factor Q is applied once. The phase residual is

$$R_\phi = \phi_t - M(T)[\varepsilon^2 \Delta \phi - 2B\phi(1 - \phi)(1 - 2\phi) - 6D(T_c - T)\phi(1 - \phi)]. \quad (16)$$

Thus the thermal interface is a spatial cell-balance residual and the phase interface is a pointwise AD residual. They are not claimed to equal the previous pointwise thermal formulation or the full space-time construction of [6]. At $U = 0$, E uses the analytic electrical solution $v = q = 0$ while thermal and phase evolution remain active.

3.3 Soft comparator and training objectives

The soft electrical PINN, F, trains all three field heads. Its potential is $V_\theta = U(t) \exp(h_v)/[\exp(h_v)+1-z]$, preserving the voltage range and the exact top value. It evaluates the same face operator and Joule deposition as E, but penalizes

$$r_{e,i} = (A(\sigma_\theta)V_\theta - f(U, \sigma_\theta))_i/\omega_i. \quad (17)$$

No additional, differently averaged electric boundary loss is added: electrical boundary conductances already enter this residual. Let L_{obs} denote the common weighted observation loss, including the full initial-logit increment for phase, and let L_{BC} and L_{IC} be the common remaining boundary and initial losses. Their exact measures are specified in Supplement S2. With one calibration a, b at the shared parent and λ rising to 0.1,

$$J_{T\phi} = \frac{1}{3} \mathbb{E}_\rho[(R_T/4)^2 + (R_\phi/5)^2], \quad (18)$$

$$\mathcal{L}_E = L_{\text{obs}, E}/a + \lambda(5L_{\text{BC}} + L_{\text{IC}} + J_{T\phi, E})/b, \quad (19)$$

$$\mathcal{L}_F = L_{\text{obs}, F}/a + \lambda(5L_{\text{BC}} + L_{\text{IC}} + J_{T\phi, F})/b + \frac{\lambda\eta}{3b} \mathbb{E}_\rho[r_e^2]. \quad (20)$$

Here ρ is the uniform space-time target with the declared time-window quadrature. Scales a and b are shared within a pair and frozen, with a numerical floor of 10^{-12} . The primary F uses $\eta = 1$. A development control uses a one-time gradient-balanced η ; a second control replaces the 128 sampled electrical cells by the volume mean over all 3200 cells at each physical time. The latter changes spatial integration, not the penalty by a factor of 25. Thermal and phase terms keep their original interfaces.

E freezes the unused potential head. Its voltage observations differentiate through conductivity and the solve, whereas F voltage observations act on its potential head. E enforces its constraint at powered observation and physics times; F penalizes it at physical residual times. Initial potential, active parameters, constraint strength and gradient maps consequently differ. These are explicit parts of the method-package comparison, not hidden evidence for an isolated VJP mechanism.

3.4 The D_E ablation and strong interpolant

E learns T and ϕ through the electrical solve and includes the thermal/phase interior residuals, boundary terms and initial constraints. F retains a neural potential and the frozen finite electrical penalty; its endpoint receives the identical electrical readout used by E. D_E shares E's neural state mapping, temperature adapter, electrical solve, complete first-order conductivity feedback, voltage-observation loss, boundary/initial constraints, parent, calibration and optimization limits; only the thermal/phase interior package is removed. D_E is consequently a physics-containing ablation, not a data-only network.

B_E uses time-PCHIP and spatial-linear interpolation of the supplied T/ϕ observations, the original initial-logit restoration and known boundary extensions, followed by the same electrical solve. It is constructed once per protocol/information condition and does not use the V labels. The E/ B_E comparison therefore measures practical competitiveness under the stated input utilization, not a strictly identical information-use intervention.

The precise ablation objective is

$$\mathcal{L}_{D_E} = L_{\text{obs}, E}/a + \lambda(5L_{\text{BC}} + L_{\text{IC}})/b. \quad (21)$$

The scale b is still calibrated from the complete parent physical package. Thus E minus D_E removes exactly the interior term rather than renormalizing the shared boundary or observation contribution. Targeted value/gradient checks and full-visible compatibility checks precede the new execution.

3.5 Common parents, fixed endpoints and readouts

For each protocol, seeds 29 and 43 initialize fresh networks and a zero-output adapter, without loading trained weights, optimizer history or calibration. A common observation-only parent receives 2400 Adam updates [9] and 600 full-observation L-BFGS evaluations, 200 per field head [10]. E and F then start from that same fitted parent with fresh optimizers. Each branch receives 1500 Adam updates and at most 300 fixed complete-objective/gradient evaluations. Every line-search trial and repeated evaluation counts; budget termination restores the last accepted state. No reference score selects a checkpoint.

The learning rate, Adam parameters, clipping, observation stream, physical time windows and residual scales are shared. L-BFGS uses fixed data and quadrature, without resampling or gradient clipping inside its closure. Equal update and evaluation caps do not imply equal computation: E incurs sparse forward and adjoint solves, while F differentiates a potential network and its explicit residual. Actual work is retained in Supplement S6; no speedup is claimed.

The shorter-protocol D_E controls share the corresponding full-label parents. B1 instead establishes two entirely new observation-only parents with only its visible phase labels, then runs E , D_E and F in each seed without selecting a favorable checkpoint or skipping an arm based on performance. All fixed endpoints are locked before reference scoring. The common 160 by 80 and 240 by 120 electrical readers evaluate every model at the same 1001 times. Fine-grid T/phase are direct queries of the same locked function. T/phase and event scores remain on 160 by 80; only fine potential is conservatively restricted for the inherited field guard, while native currents, power and Joule deposition stay native.

3.6 Distinct error and event measures

The historical primary comparison uses the same 160×80 electrical layer at the reference times. The 240×120 reader changes prediction-side electrical discretization while preserving the historical scoring measure. The unprojected F readout is retained separately. For a fixed F , projection changes V , q and electrical readouts while leaving T , φ and every phase event unchanged. The error measures distinguish these effects.

S is the space-time mean symmetric difference of active sets, where active means $\varphi \geq 0.5$; it uses the full cell domain and global trapezoidal time weights. Raw E_φ is phase RMS on the ROI $|x| \leq 0.55$, $0 \leq z \leq 0.55$. E_T is ROI temperature RMS divided by 0.45, and E_V is unnormalized full-domain potential RMS. Device errors are

$$\mathcal{E}_I = \frac{\|I - I_{\text{ref}}\|_t}{\|I_{\text{ref}}\|_t}, \quad \mathcal{E}_P = \frac{\|P_J - P_{J, \text{ref}}\|_t}{\|P_{J, \text{ref}}\|_t}, \quad \mathcal{E}_W = \frac{|\int (P_J - P_{J, \text{ref}}) dt|}{|\int P_{J, \text{ref}} dt|}. \quad (22)$$

The time norm is trapezoidal RMS over the complete interval. The matched device criterion compares predicted bottom current with the reference top-current trace, retaining the

historical scoring convention. The top-current noninferiority guard uses the predicted top current. For projected states these terminal traces agree to solve accuracy; the archive also reports bottom-native errors as a separate diagnostic. Unprojected soft currents can disagree substantially and must not be interchanged. Percentages below are 100 times normalized errors, not dimensional percentage errors in temperature or phase. All absolute, percentage-point and relative differences are provided in the unified tables.

For the same heating-window measure, let M_{pred} , M_{ref} and M_{TP} be predicted, reference and overlapping active masses. When both masses exceed the numerical denominator floor, the definitions give

$$\text{recall} = \text{precision} \frac{M_{\text{pred}}}{M_{\text{ref}}} \leq \min \left(1, \frac{M_{\text{pred}}}{M_{\text{ref}}} \right). \quad (23)$$

This identity separates an insufficient predicted active mass from mismatch in space and time at a fixed mass. It is a descriptive support diagnostic, not a new loss or evidence that a particular neural mechanism caused the error. The original near-zero floors remain in the scorer. The strict gates are project-defined evaluation requirements, not experimental device-reliability standards or estimates of a success probability.

Predeclared phase-reconstruction and device-function criteria require 10% improvements in their named primary metrics with 5% noninferiority guards and fixed absolute floors. They are retained in Supplement S3, rather than used as a substitute for reporting error magnitudes. Failure to cross an advantage threshold does not demonstrate equivalence. Strict event usability is separate: both cycles must meet onset error ≤ 0.005 , recall ≥ 0.9 , precision ≥ 0.8 , mass ratio 0.8-1.2, and the specified peak, locality and recovery conditions. An onset is the linearly interpolated upward crossing of ROI active fraction 0.02. Support metrics integrate cell-volume overlap during each heating window.

For projected states, $P_j = \text{UI}$ to electrical solve accuracy. Current and power errors thus differ in drive-dependent temporal weighting, but are not independent physical replications. Likewise, a small integrated energy error may reflect cancellation. We therefore retain the signed per-pulse errors, event metrics and local state errors even when global device criteria pass.

3.7 Frozen second-cycle phase-gap experiment

The new observation condition removes phase labels on the complete second cycle $W=[1.01,2.02]$ of the existing shorter protocol. V/T observations and phase outside W , including $t>2.02$, remain available. The exported data contain 17,094 positive-time phase labels and omit 11,781; V and T each retain 28,875. The original observed grid contains 1.02 through 2.02 within W , with no observed node at 1.01. The last preceding and first following visible times are 1.00 and 2.04. The analytic initial condition remains known.

Two new observation-only parents, seeds 29 and 43, each initialize E , D_E and F under the frozen recipe. No complete-label trained state, calibration or interface pool is reused. Phase observation mass restricts the original coordinate quadrature before normalization. A threshold-straddling cell is admitted only when all its original adjacent corners remain visible. The primary question is whether E adds information over D_E in W , under the separately named A_w criterion. Whole-history A/B, strict events, outside-window costs and $E/F/B_E$ comparisons remain separate. This is offline reconstruction of a developed protocol, not online forecasting or a wholly unobserved phase trajectory.

A_w applies the original phase-advantage formula to W with its own time weights and normalizers: the S and phase-RMS improvements must meet the inherited 10% or

absolute-floor requirements, while T, top-current and potential retain their original 5% noninferiority guards. The bottom-current convention remains separately reported. The outside measure integrates [0,1.01] and [2.02,2.5] separately, without bridging W. Its unnormalized integrals plus those of W reproduce the full-history integrals. Outside costs include all seven named state/device quantities. Original full-history A/B and strict-event rules remain unchanged. This is a complete withheld time window for phase, with V/T still observed inside it; it is not a formal out-of-distribution or forecasting test.

4. Reconstruction after common electrical repair

4.1 Four paired full-label comparisons

All four clean E/F pairs retain the complete device criterion under every existing reference and both readers. Under the spatial reference with the finer reader, current-error reductions are 47.03-76.52% (0.78-2.28 percentage points) and power-error reductions are 47.86-77.50% (0.82-2.39 points). The comparison therefore survives giving F the same final electrical solve. Four pairs on two related protocols are the comparison units; reference, reader and saved-time counts do not increase that sample size.

Table 1. Complete full-label objects under the spatial reference and fine reader. Phase uses the original 160 by 80 ROI; native device NRMSE values are percentages. B_E is one deterministic object per protocol. D_E is shown alongside the learned controls rather than introduced only after the main results.

Protocol	Seed	Role	Phase RMS	Current (%)	Power (%)
original	29	E	0.0210621	0.883264	0.889983
original	29	F	0.0247498	1.66733	1.707
original	43	E	0.0224575	1.59785	1.62156
original	43	F	0.0251648	3.07484	3.18207
original	shared	B_E	0.0233096	1.61121	1.63659
shorter	29	E	0.0199363	1.01558	1.0237
shorter	29	F	0.0244496	2.27681	2.34323
shorter	43	E	0.0197631	0.70046	0.694693
shorter	43	F	0.0240993	2.98378	3.08711
shorter	shared	B_E	0.0252168	1.9043	1.9475
shorter	29	D_E	0.0194683	0.958122	0.965194
shorter	43	D_E	0.0197085	0.642947	0.624503

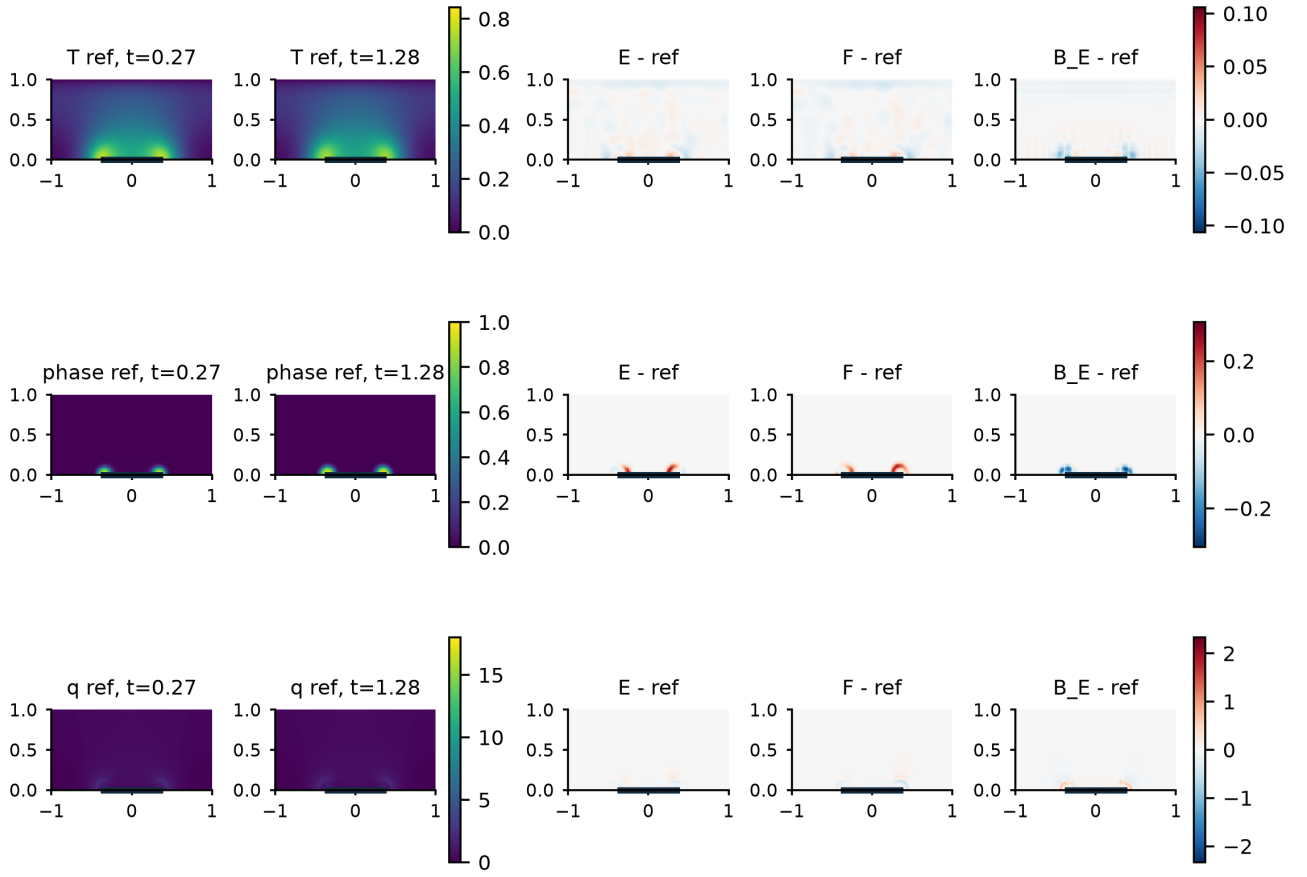


Figure 2. Native 240 by 120 physical states and signed errors for the full-label shorter protocol. The first two columns show spatial-reference T , phase and Joule density at the two predeclared plateau endpoints. The remaining columns show E , repaired F and B_E minus reference at $t=1.28$ for seed 29. Error color scales are shared with the seed-43 companion in the supplement. These physical panels are descriptive native-grid views; the primary T /phase/event measure stays on 160 by 80.

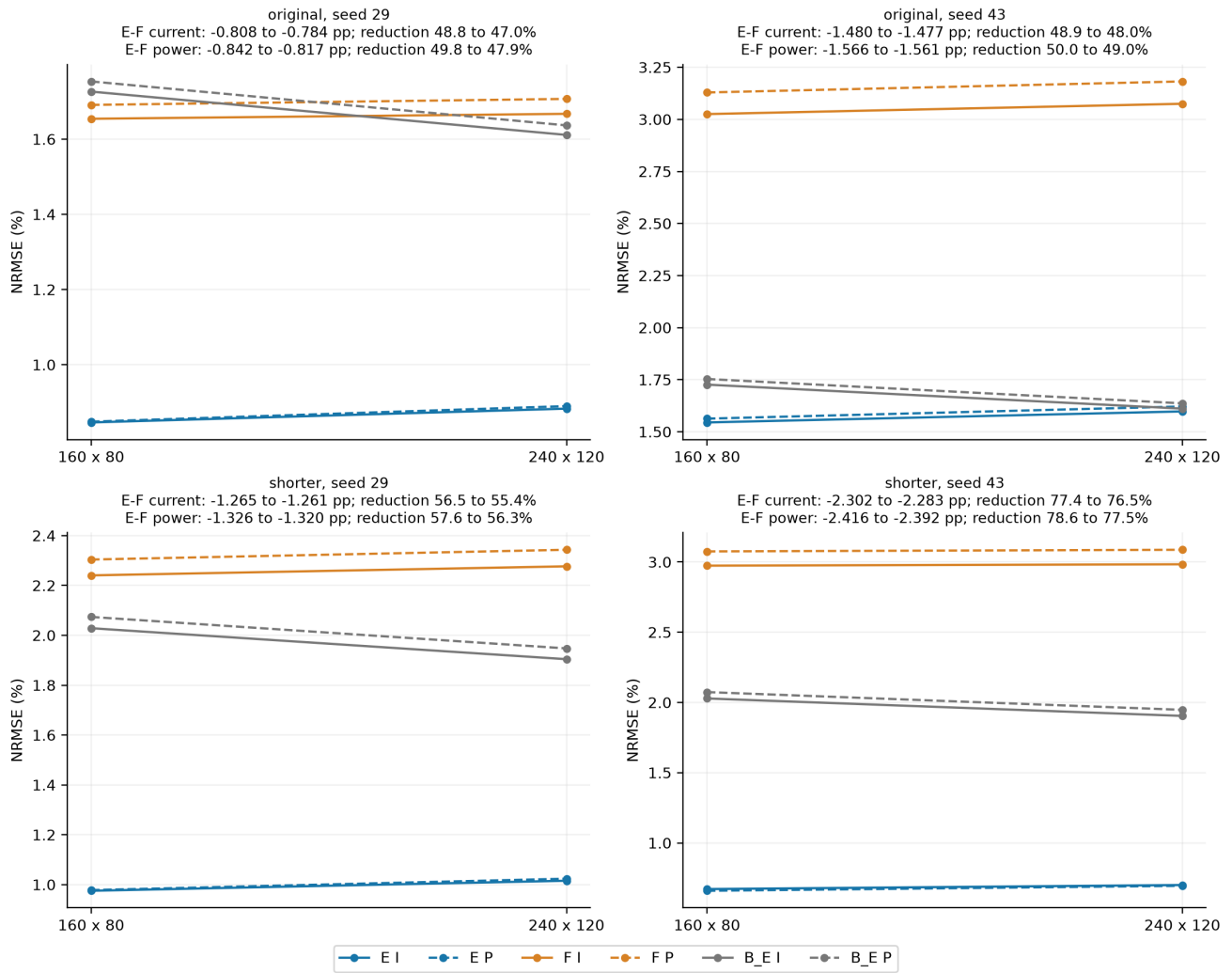


Figure 3. All four historical E/F pairs under the same native spatial reference and the two electrical readers. Solid and dashed curves show current and power NRMSE. B_E is repeated visually across seeds but is one baseline per protocol. Signed E-F differences use percentage points. The complete criterion also includes all field guards.

The development controls illustrate why repair is a necessary comparison: with T/phase fixed, projection reduces F_{raw} current NRMSE from 279.30% to 1.071% and power NRMSE from 20.118% to 1.065%, while E gives 0.694% and 0.682%. A full-spatial finite-penalty control does not remove that development difference, but its integrated energy error is better. These controls delimit the tested package; they are retained in Supplement S4 and are not pooled as clean replications.

4.2 The strong interpolation counterexample

This robustness does not extend uniformly to B_E . For original-protocol seed 43 and the 240×120 reader, the full-precision records give:

Table 2. Continuous interpolation counterexample for original-protocol seed 43 with the fine reader. Negative relative error reduction means larger E error; full-precision records determine the unchanged A/B decisions.

Reference	Current NRMSE: E / B_E (%)	Power NRMSE: E / B_E (%)	Relative current / power error reduction (%)	Original A / B
Original	1.631055 / 1.620006	1.657409 / 1.646429	-0.681976 / -0.666882	False / False

Reference	Current NRMSE: E / B _E (%)	Power NRMSE: E / B _E (%)	Relative current / power error reduction (%)	Original A / B
Time-refined	1.653271 / 1.570460	1.681856 / 1.596088	-5.273031 / -5.373634	False / False
Space-refined	1.597848 / 1.611207	1.621556 / 1.636592	0.829142 / 0.918751	False / False

Negative reduction denotes larger E error. Under the time-refined reference the differences are 0.082811 and 0.085768 percentage points, respectively. Under the spatial reference E remains slightly better, below the prescribed practical margin. The first two comparisons already failed B, so the continuous reversals introduce no additional Boolean decision changes. Full-precision source fields and line locations are preserved in [the P04 table](#); these display values do not re-adjudicate thresholds.

5. Additional dynamic residuals: full-label and phase-gap tests

5.1 Full-label matched D_E evidence

In both clean shorter-protocol pairs, D_E has lower phase, current and power RMS errors than E under all three references and both readers. Neither pair establishes the prescribed added-residual phase or device increment. With the spatial reference and finer reader, E exceeds D_E in phase/current/power error by 2.404%/5.997%/6.061% for seed 29 and 0.277%/8.945%/11.239% for seed 43. Other outcomes are mixed. Because the electrical solve and voltage feedback remain active in D_E, this result concerns the additional thermal/phase interior package. It does not test physics against a data-only network. Complete independent audit components remain in Supplement S17.

5.2 Second-cycle phase-gap reconstruction

Neither initialization establishes the complete phase-gap criterion against D_E under the three references and two readers. Across those conditions, relative phase-RMS reductions of E against D_E range from -4.23% to 1.23%; negative reduction means larger E error. The prescribed phase-gap experiment does not establish the claimed additional residual increment; individual favorable errors do not replace the complete criterion.

Seed 29 has phase-RMS error reductions of 1.10% to 1.23% and active-set error reductions of 1.30% to 1.46%. Seed 43 has phase-RMS error reductions of -4.23% to -4.18% and active-set error reductions of -5.25% to -5.21%. All three window guards jointly pass in 12/12 conditions. The phase-gain requirements still determine the complete A_w outcome; negative reductions indicate larger E errors.

Table 3. B1 errors inside W under the spatial reference and fine reader. All seven objects are included. Full-precision values for all references/readers, including their own denominators, remain in the complete tables.

Seed	Role	1000 S	Phase RMS	T (%)	Current (%)	Power (%)
29	E	8.92423	0.102408	1.14001	1.10488	1.11993
29	D _E	9.042	0.103548	1.13479	1.15561	1.17082
29	F	7.75023	0.121882	1.16888	8.96009	8.42207
43	E	8.0082	0.0906255	1.10747	0.931022	0.95043
43	D _E	7.61158	0.0869874	1.12689	0.996119	1.00989

Seed	Role	1000 S	Phase RMS	T (%)	Current (%)	Power (%)
43	F	7.75023	0.109148	1.11942	8.90024	8.38301
-	B _E	7.75023	0.127808	1.41376	9.08846	8.55694

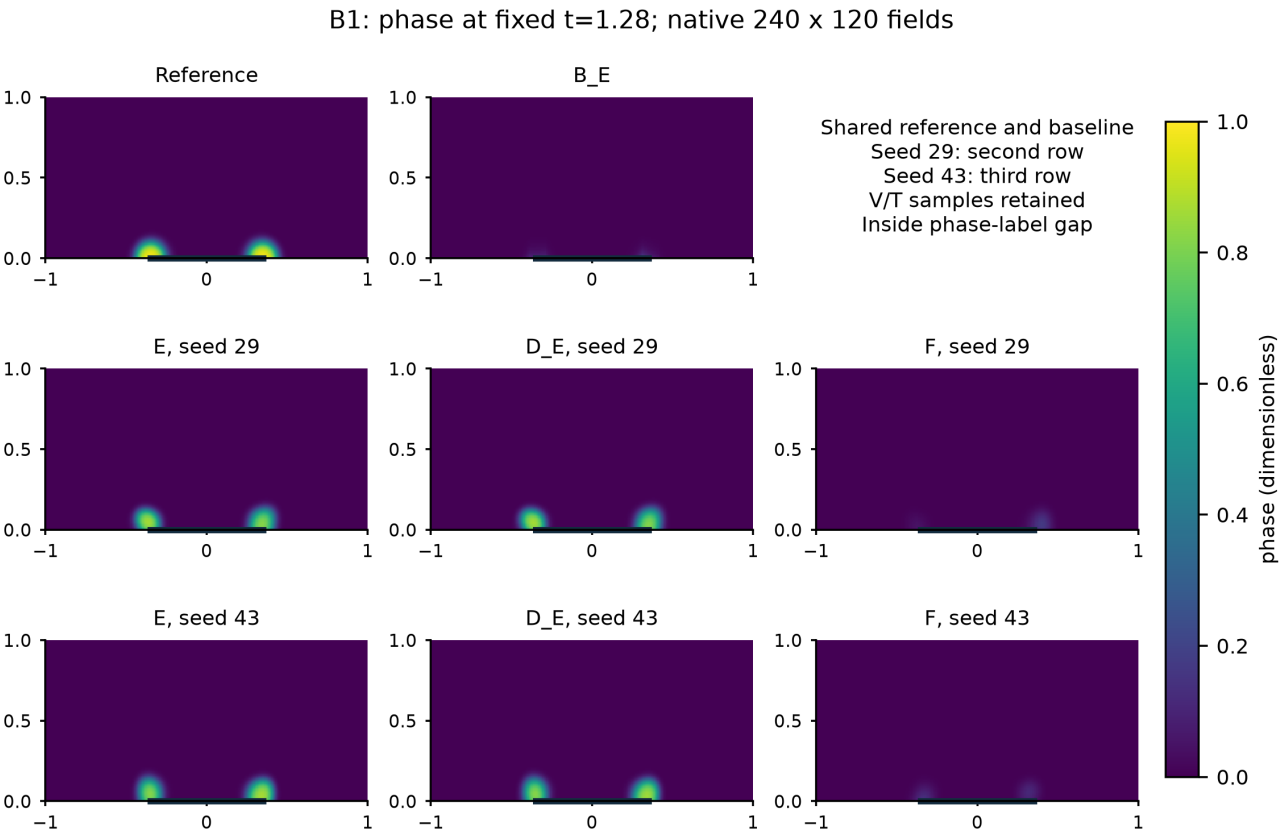


Figure 4. B1 phase at t=1.28, inside the withheld second cycle. Reference and shared B_E accompany E, D_E and F for both seeds. All panels use the same [0,1] color range. Native fine-grid fields provide a physical view, while the primary state and event criteria use the original coarse measure. T, local q, both predeclared times and signed-error companions are retained in the supplementary physical atlas.

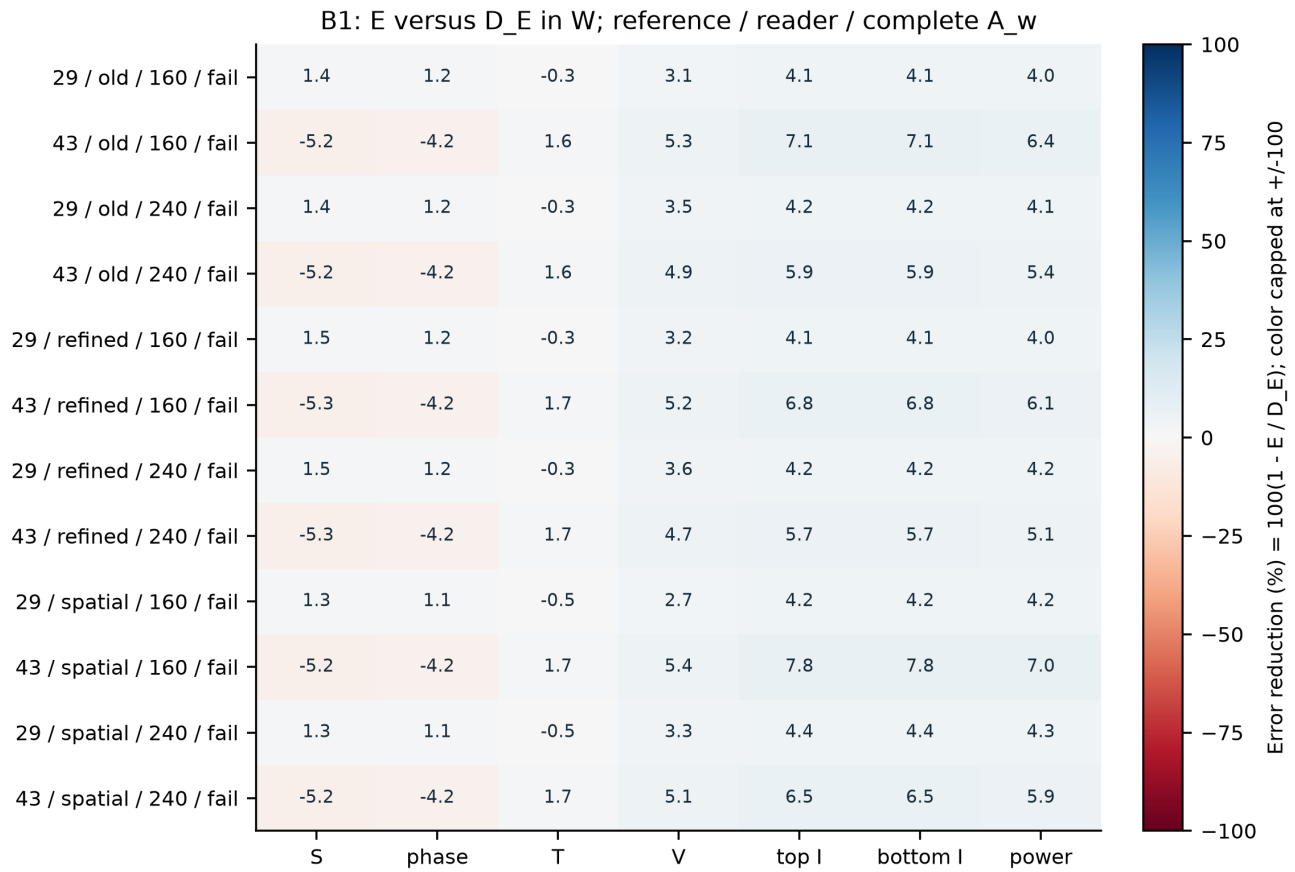


Figure 5. Window effects of E against same-parent D_E in all twelve seed/reference/reader conditions. Positive numbers mean smaller E error. Colors are capped at plus/minus 100% for legibility; annotations retain the actual values. The row label gives the complete A_w outcome, which also uses absolute floors and noninferiority. The rows are sensitivity checks on two independent initializations, not twelve independent experiments.

5.3 Whole-history costs and strict events

Outside-window noninferiority costs occur in 6 of the twelve primary comparison conditions. The affected quantities are top-current NRMSE, potential RMS, bottom-current NRMSE, power-trace NRMSE, with every condition retained in the comparison table. All seven B1 objects fail the complete strict two-cycle rule under every tested reference and reader.

Table 4. Primary B1 decisions in every reference/reader condition. Full A/B retain their original definitions; outside cost denotes at least one separately reported noninferiority violation. A favorable window result does not erase an outside or strict-event cost.

Reference	Reader	Seed	A _w	Full A	Full B	Outside cost
old	coarse	29	False	False	False	False
old	coarse	43	False	False	False	True
old	fine	29	False	False	False	False
old	fine	43	False	False	False	True
refined	coarse	29	False	False	False	False
refined	coarse	43	False	False	False	True
refined	fine	29	False	False	False	False
refined	fine	43	False	False	False	True

Reference	Reader	Seed	A_w	Full A	Full B	Outside cost
spatial	coarse	29	False	False	False	False
spatial	coarse	43	False	False	False	True
spatial	fine	29	False	False	False	False
spatial	fine	43	False	False	False	True

Against the repaired soft PINN F, B1 E satisfies the same window reconstruction formula in 0/12 conditions and the original whole-history A/B rules in 0/12 and 6/12, respectively. Against the shared interpolant B_E , B1 E satisfies the same window reconstruction formula in 0/12 conditions and the original whole-history A/B rules in 0/12 and 10/12, respectively. These are reference/reader checks on the same two initializations. Their continuous effects and all failures remain in the complete tables; the primary attribution of the additional interior package still uses E versus D_E . For seed 29, the full device rule passes 0/6 checks against F and 4/6 against B_E . For seed 43, the full device rule passes 6/6 checks against F and 6/6 against B_E .

All E/F and E/ B_E comparisons, complete cycle metrics, signed device effects and separate full/outside normalizers are retained with the primary comparison. Phase observations after 2.02 support both the networks and the legal interpolant, so even a favorable result is offline reconstruction rather than online second-pulse prediction. The outcome has not changed the full-label D_E finding, historical E/F decisions, seeds, thresholds or reference definitions.

6. Physical interpretation and numerical scope

6.1 Pulse history and functional quantities

The references agree before the changed pulse to roundoff at matching resolution. Immediately before the second pulse, advancing it from 1.25 to 1.01 increases the reference ROI mean T from 0.006525 to 0.018746 and mean phase from 0.0004829 to 0.0039725. The second-event latency decreases from 0.2484 to 0.2168. These observations support history dependence in this numerical object but do not isolate thermal from phase mediation. Complete per-pulse states and event values remain in Supplement S5.

Device and event rankings differ. Historical shorter-protocol E passes both onset-timing tolerances under the original reference, but first-cycle recalls 0.866607 and 0.875507 remain below 0.9. F29 has the smaller second-cycle timing error. F43 has opposite signed pulse-energy errors, +0.00511674 and -0.00400099, yielding 0.25736% total energy error despite 3.09469% power-trajectory NRMSE. E also exhibits cancellation. Since projected current and power obey $P=UI$, their errors are related drive-weighted views; neither low energy error nor electrical balance certifies local heating or strict event reconstruction.

6.2 Reference and reader sensitivity

The original reference uses 160 by 80 cells and $dt=0.000625$; temporal refinement halves dt ; spatial refinement uses 240 by 120 at the halved step. All save the same 1001 times and keep the learned states fixed. Spatial fields are conservatively restricted for the primary state/event measure while ports remain native. Predictions have two separately tested electrical readers. These are finite perturbation tests, not a demonstrated convergence order or continuum-error estimate.

Across the historical coarse-reader comparisons, temporal refinement changes no A/B decision; spatial refinement changes three: original/29 E/B_E loses A, shorter/43 E/F loses A, and original/43 E/B_E gains B. The latter B crossing disappears with the finer reader. Section 4.2 additionally retains the continuous B_E reversals under original/time references. The complete E/F device criterion remains retained in every tested condition. Supplement S9 and S14-S16 give the complete margins, sufficient perturbation bounds and restriction conventions.

A later gated continuation crosses the strict rule only under the time-refined reference and loses it under spatial refinement as first-cycle recall changes from 0.902063024 to 0.897526906. Ordinary and gated phase-head extensions do not establish a stable matched increment over unchanged continuation. All six negative continuation states are retained in Supplement S10-S13 and the separately indexed array extension. These limits prevent attributing the core effect to a demonstrated new gate, extra capacity or arbitrary-grid robustness.

6.3 What the experiment identifies

The E/F contrast identifies the tested constraint-training configuration. Initial V, active parameters, exact versus finite enforcement, enforcement times and gradient maps differ; the experiment does not isolate one implicit-gradient pathway. E/D_E controls that distinction more tightly for the additional interior residuals, but its conclusion is conditional on the information regime and frozen optimization recipe. Each information regime rebuilds its own parents and calibration scales. The comparison between regimes therefore describes the observed pattern of increments without holding every training mediator fixed. Equal update limits do not give equal numerical work, and no acceleration over a conventional forward solver is claimed.

The domain has a localized electrode and a closed electric-thermal-phase feedback chain. Its dimensionless coefficients, single phase variable and idealized geometry still require a separate material-consistent calibration and independent validation before a device-material claim. The initial-boundary incompatibility and shared numerical discretization likewise remain limitations. A stronger algorithmic or material claim would need a separately designed experiment rather than relabeling the present controls.

7. Conclusions and availability

Training-time electrical elimination with consistent local Joule deposition produces a matched device-reconstruction benefit over the tested soft PINN after both receive the same final electrical repair. All four full-label pairs retain the complete device criterion under the declared reference and reader changes. Strong interpolation supplies a continuous reversal and a reader-sensitive advantage decision, limiting the scope of that benefit.

The full-label E/D_E evidence does not establish added predictive value from the interior thermal/phase residuals. Neither initialization establishes the complete phase-gap criterion against D_E under the three references and two readers. The prescribed phase-gap experiment does not establish the claimed additional residual increment; individual favorable errors do not replace the complete criterion. Outside-window noninferiority costs occur in 6 of the twelve primary comparison conditions. The affected quantities are top-current NRMSE, potential RMS, bottom-current NRMSE, power-trace NRMSE, with every condition retained in the comparison table. All seven B1 objects fail the complete strict two-cycle rule under every tested reference and reader. The work separates

configuration-level evidence, tests of additional residual value and unresolved material or generalization claims.

Data, code and author declarations

The curated repository at [commit 218bb66069da52b2ccfe9dd68ac519edbe4584d0](https://commit.218bb66069da52b2ccfe9dd68ac519edbe4584d0) contains the 18 September scientific revision and selected evidence, published separately on 20 September 2026. The present B1 revision remains local. Its portable array package contains 72 full-label core records, a separately preserved 18-record negative continuation extension and 42 B1 reference/reader records. Those record counts are not independent-repetition counts. The core reproduced 8,868 saved scalar/identity/Boolean entries at $\text{rtol}=2\text{e-}10$ and $\text{atol}=2\text{e-}12$, with exact categorical agreement. B1 was first scored from its separately portable NumPy-only inputs after endpoints and readers were locked and GPU recovery/shutdown completed. Array arithmetic, figure/PDF rebuilding, new model inference and retraining are distinct reproduction levels. No independent external retraining is claimed.

Full arrays have not been publicly uploaded or assigned a DOI. Public or controlled reviewer access, appropriate distribution permission and persistent archiving require author approval. AI-assisted preparation: OpenAI Codex assisted with research-code preparation, numerical execution, source checking, analysis, figures and drafting. Human authors must verify the content and take responsibility for interpretation, disclosures and submission. Funding, author contributions, competing interests, affiliations and final author approval remain to be supplied truthfully. This numerical study contains no experimental or human-subject data.

References

- [1] Raissi, M., Perdikaris, P., and Karniadakis, G. E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* 378 (2019), 686-707. DOI: [10.1016/j.jcp.2018.10.045](https://doi.org/10.1016/j.jcp.2018.10.045). [Author preprint, Part I](#).
- [2] Wang, S., Teng, Y., and Perdikaris, P. Understanding and mitigating gradient flow pathologies in physics-informed neural networks. *SIAM Journal on Scientific Computing* 43(5) (2021), A3055-A3081. [Author full text](#).
- [3] Um, K., Brand, R., Fei, Y. R., Holl, P., and Thuerey, N. Solver-in-the-Loop: Learning from Differentiable Physics to Interact with Iterative PDE-Solvers. *Advances in Neural Information Processing Systems* 33 (2020). [Author full text](#). [Official project](#).
- [4] Mitusch, S. K., Funke, S. W., and Kuchta, M. Hybrid FEM-NN models: Combining artificial neural networks with the finite element method. *Journal of Computational Physics* 446 (2021), 110651. DOI: [10.1016/j.jcp.2021.110651](https://doi.org/10.1016/j.jcp.2021.110651). [Author full text](#).
- [5] Blondel, M., Berthet, Q., Cuturi, M., Frostig, R., Hoyer, S., Llinares-López, F., Pedregosa, F., and Vert, J.-P. Efficient and Modular Implicit Differentiation. *Advances in Neural Information Processing Systems* 35 (2022). [Author full text](#).
- [6] Patel, R. G., Manickam, I., Trask, N. A., Wood, M. A., Lee, M., Tomas, I., and Cyr, E. C. Thermodynamically consistent physics-informed neural networks for hyperbolic systems. *Journal of Computational Physics* 449 (2022), 110754. DOI: [10.1016/j.jcp.2021.110754](https://doi.org/10.1016/j.jcp.2021.110754). [Author full text](#).
- [7] Allen, S. M., and Cahn, J. W. A microscopic theory for antiphase boundary motion and its application to antiphase domain coarsening. *Acta Metallurgica* 27(6) (1979), 1085-1095. DOI: [10.1016/0001-6160\(79\)90196-2](https://doi.org/10.1016/0001-6160(79)90196-2).
- [8] Miquel, R., Cabout, T., Cueto, O., Sklénard, B., and Plapp, M. Multi-physics modeling of phase change memory operations in Ge-rich Ge₂Sb₂Te₅ alloys. *Journal of Applied Physics* 136 (2024), 145102. DOI: [10.1063/5.0222379](https://doi.org/10.1063/5.0222379). [Author full text](#).

- [9] Kingma, D. P., and Ba, J. Adam: A Method for Stochastic Optimization. International Conference on Learning Representations (2015). [Author preprint](#).
- [10] Liu, D. C., and Nocedal, J. On the limited memory BFGS method for large scale optimization. Mathematical Programming 45 (1989), 503-528. [Author publication page and paper](#).
- [11] He, K., Zhang, X., Ren, S., and Sun, J. Deep Residual Learning for Image Recognition. IEEE Conference on Computer Vision and Pattern Recognition (2016), 770-778. [Author preprint](#).
- [12] Baez, A., Zhang, W., Ma, Z., Nguyen, L., Das, S., and Daniel, L. Guaranteeing Conservation of Integrals with Projection in Physics-Informed Neural Networks. arXiv:2511.09048v2 (2026). [Author record](#). [Author full text](#).
- [13] Kaltenbacher, B. Regularization based on all-at-once formulations for inverse problems. SIAM Journal on Numerical Analysis 54 (2016), 2594-2618. [Author preprint](#). DOI: [10.1137/16M1060984](#).
- [14] Shi, Y., and Chen, L.-Q. Isothermal current-driven insulator-to-metal transition in VO₂ through strong correlation effect. Physical Review Applied 11 (2019), 014059. [Author preprint](#). DOI: [10.1103/PhysRevApplied.11.014059](#).
- [15] Négier, G., Mahoney, M. W., and Krishnapriyan, A. S. Learning differentiable solvers for systems with hard constraints. International Conference on Learning Representations (2023). [Author record](#). [Author full text](#).
- [16] Lu, L., Pestourie, R., Yao, W., Wang, Z., Verdugo, F., and Johnson, S. G. Physics-informed neural networks with hard constraints for inverse design. SIAM Journal on Scientific Computing 43(6) (2021), B1105-B1132. DOI: [10.1137/21M1397908](#). [Author full text](#). [Official implementation](#).
- [17] Golder, R., Roy, B. N., and Hasan, M. M. F. DAE-HardNet: A Physics Constrained Neural Network Enforcing Differential-Algebraic Hard Constraints. arXiv:2512.05881v1 (5 December 2025), preprint. [Author record](#).
- [18] Horne, M. J. S., Jimack, P. K., Khan, A., and Wang, H. Hard Constraint Projection in a Physics Informed Neural Network. ParCFD2024 conference paper; publication record 2025, DOI: [10.34734/FZJ-2025-02453](#). Accepted manuscript archived as arXiv:2601.06244v1 (9 January 2026). [Author record](#). [Publisher metadata](#).